

A construction-dependent extension of EurOtop for predicting wave overtopping during rubble-mound breakwater construction

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Abstract

Wave overtopping represents a significant hazard during the construction of rubble-mound breakwaters (Diaz-Hernandez et al. 2023, 2017). While commonly used overtopping formulations, such as those included in EurOtop manual, provide reliable estimates for completed structures, their applicability to intermediate construction stages remains limited. During construction, changes in crest freeboard, exposed layers, roughness and permeability can substantially modify the hydraulic response of the structure. Consequently, overtopping may be significantly underestimated, potentially compromising worker and equipment safety and causing damage, construction delays and additional costs.

Method and results

This study investigates overtopping during different construction stages of rubble-mound breakwaters and proposes a construction-dependent extension of the EurOtop formulation (Van der Meer et al., 2018). A systematic set of two-dimensional numerical experiments was performed using the IH2VOF Volume of Fluid (VOF) model (Losada et al. 2008). A simplified rubble-mound breakwater composed of armour and two underlayers was progressively modified to reproduce different construction stages. A Design of Experiments (DoE) approach was used to select physically relevant combinations of wave conditions, resulting in 125 numerical simulations covering significant wave heights from 2 to 12 m and wave steepness values from 0.015 to 0.070.

The numerical experiments reveal a strong dependence of overtopping discharge on the degree of completion of the structure. For the configurations analysed, a breakwater approximately 90% complete can experience overtopping discharges up to one order of magnitude larger than conventional EurOtop predictions, while differences exceeding two orders of magnitude were obtained for approximately 50% completion. These results demonstrate that explicitly accounting for the construction stage can be essential when assessing overtopping hazards during construction.

To address this limitation, a new Construction Completion Index is introduced to provide a compact representation of the hydraulic state of the partially constructed breakwater. The index combines the residual emerged height of the structure with the degree of completion of the armour and underlayers through calibrated weighting factors. A construction-dependent correction coefficient is subsequently incorporated into the EurOtop formulation. Its parameters are obtained through an optimisation procedure and logarithmic regression against the numerical results. For the calibration dataset (see Fig. 1), the proposed formulation reproduces the simulated mean overtopping discharges across the different construction stages with an RMSE of approximately 12% and an R^2 of 0.94, indicating a satisfactory overall agreement between the numerical results and the modified EurOtop predictions.

$$q = \sqrt{g H_{m0}^3} \cdot 0.09 \cdot \exp\left(-\left(\frac{1.5 R_c}{H_{m0} \gamma_f K_c}\right)^{1.3}\right)$$

$$I_h = w_R I_R + w_A I_A + w_{U1} I_{U1} + w_{U2} I_{U2}$$

$$K_c = 1 + \alpha(1 - I_h)^\beta$$

I_h is the construction index, I_i is the proportions of each layer, w_i is the associated calculated weights, K_c is the construction factor and α, β are the calibrated parameters

The transferability of the methodology is further explored through additional simulations involving variations in breakwater slope and crest freeboard. The results show encouraging agreement for changes in slope, while modifications in freeboard reveal a tendency of the proposed formulation to underestimate overtopping, highlighting the current limits of extrapolation. An additional blind test using an extreme section representative of the Langosteira breakwater (Galicia, Spain) confirms the ability of the methodology to reproduce the behaviour of the later construction stages, although some discrepancies arise during the earliest stages.

Finally, the methodology is extended beyond mean overtopping discharge to characterise individual overtopping volumes and extreme events. Individual volumes extracted from the numerical simulations are described using a two-parameter Weibull distribution, and the evolution of its scale and shape parameters is investigated as a function of the Construction Completion Index. The results indicate a systematic dependence of both parameters on construction progress, suggesting that the proposed framework could ultimately provide estimates of both mean overtopping discharge and maximum individual overtopping volumes from the state of completion of the breakwater.

The proposed methodology provides a first step towards construction-stage-dependent overtopping prediction, with potential application to the assessment of working conditions and overtopping hazards during rubble-mound breakwater construction.

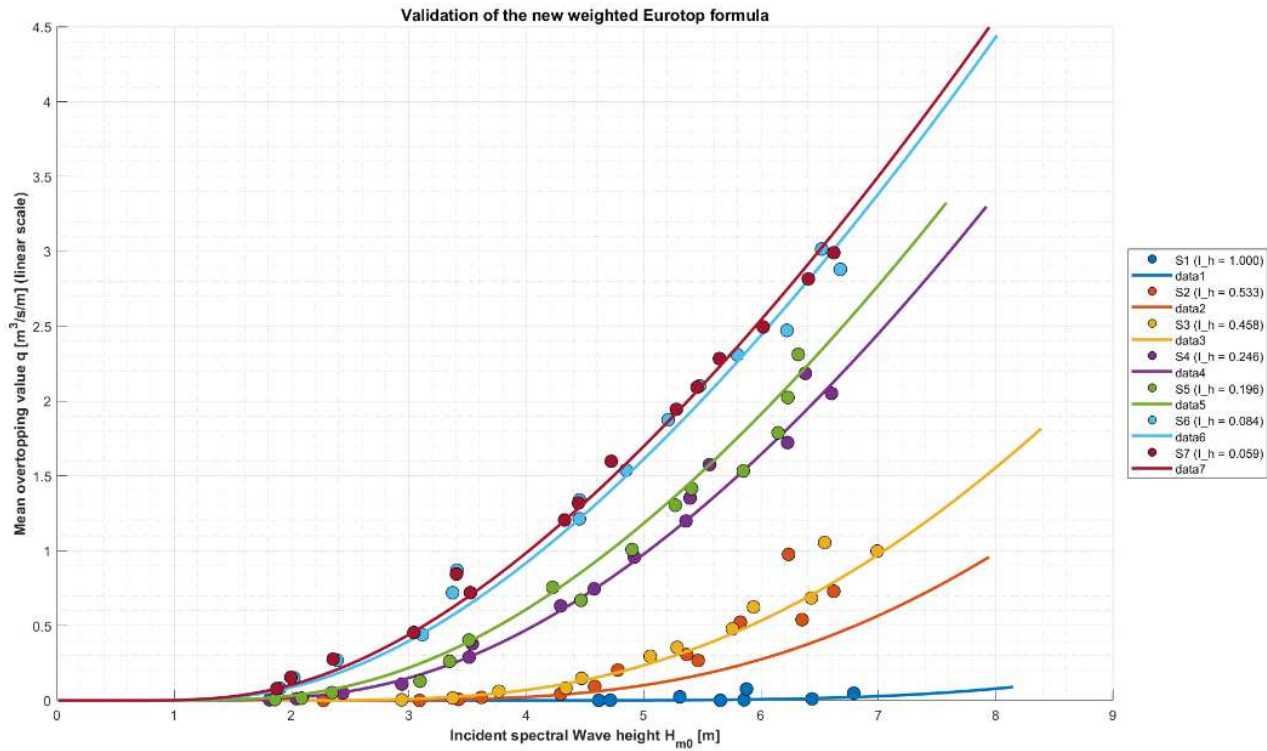


Figure 1. Validation of the proposed construction-dependent EurOtop formulation for the different breakwater construction stages. Symbols represent mean overtopping discharges obtained from the IH2VOF numerical simulations, while solid lines correspond to predictions from the modified EurOtop formulation. Each colour represents a different construction stage, characterised by its corresponding Construction Completion Index, I_h . The agreement between numerical results and predicted curves demonstrates the ability of the proposed formulation to reproduce the strong variation in overtopping discharge throughout the breakwater construction process.

References

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